Project Scenario:

Horizon Analytics: Establishing a Secure Foundation

For CST170.2 Project 2 — Building and Securing a Small-Office IT Environment

1 Business Overview

Horizon Analytics is a new data-analysis and financial-consulting start-up launching its first office in South East Queensland. The office will open with **five staff** and plans to add **two or three more employees within the next year**. The company has procured an FTTP internet service (100 Mbps/40 Mbps) and an ISP-supplied modem. The founders want a secure, scalable IT foundation that supports day-to-day business operations and protects confidential client data.

2 Staff Roles and Typical Tasks *(Context for LO1 & LO2)*

Role	**Core Activities**	**Typical Software / Data Needs**
Chief Executive & Lead Analyst	Financial modelling, large spreadsheets, secure remote access while travelling	Productivity & BI suites, remote-desktop/VPN, industry compliance tools
Data Analyst	Data cleansing and statistical processing, collaborative reports	Database & analytics software, version-control, shared storage
Marketing & Communications Coordinator	Graphic design, social-media scheduling, and content creation	Design suite, cloud marketing platforms, and moderate media storage

Role	Core Activities	Typical Software / Data Needs
Client Relations Officer	CRM updates, document sharing, and video calls	Cloud CRM, office suite, HD video-conferencing
Office Manager	General administration, bookkeeping, and staff onboarding	Accounting package, productivity tools, peripheral management

Students must determine suitable workstation components and security controls based on the duties above. No component choices are provided here.

3 Small-Office Network Requirements *(Context for LO1, LO2 & LO3)*

- **Reliable wired connectivity** for desktop PCs and peripherals.
- **Enterprise-secure Wi-Fi** for staff laptops and a segregated guest network for visitors.
- **Centralised network storage** for collaborative datasets and nightly backups.
- **Remote-access capability** for travelling staff, without exposing internal devices directly to the internet.
- **Scalability** to support three additional employees within 12 months without major re-work.
- **Regulatory expectation**: basic measures against ransomware, credential theft, and data leakage, in line with Australian SME cyber-security guidance.

Students decide the specific router/firewall, switch, access point(s), IP addressing, and VLAN layout that satisfy these needs.

4 Security Concerns Raised by Management *(Context for LO2)*

1. Increasing phishing attempts targeting financial firms.
2. Potential ransomware that could lock crucial client data.

3. Safe handling of client files when working from public Wi-Fi.
4. Protecting sensitive payroll and tax records stored onsite.

Management has asked for **actionable, business-friendly recommendations** that go beyond consumer-grade protections.

5 Budget & Growth Hints *(Optional Constraints)*

- **Estimated initial hardware budget** (PCs + core network gear + peripherals): **AUD $12 000–14 000**.
- Preference for solutions that minimise ongoing licensing fees.
- Equipment should remain useful if the staff count grows to **eight** over the next year.

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Linking This Scenario to the Project 2 Template

Learning Outcome	Evidence Students Should Provide
LO1 Components & Interactions	Identify appropriate hardware for each role; explain how selected CPU, RAM, motherboard, storage, etc. inter-operate; identify network devices (router, switch, AP).
LO2 Cyber-Security Principles	Propose and justify OS hardening measures (accounts, updates, encryption, EDR) **and** network security controls (e.g. firewall rules, secure Wi-Fi, encryption) that mitigate the four concerns listed above.
LO3 Basic Networking Concepts	Draw a labelled topology diagram, devise an IP scheme, and describe device configurations that fulfil the functional requirements and allow for future growth.

Submission Reminder

Your report must follow the **“CST170.2 – Project 2 Template”**. Use the Horizon Analytics scenario details when populating each section, but do **not** copy this document verbatim into your submission. All hardware, software, and security choices must be explained and referenced in Australian English.